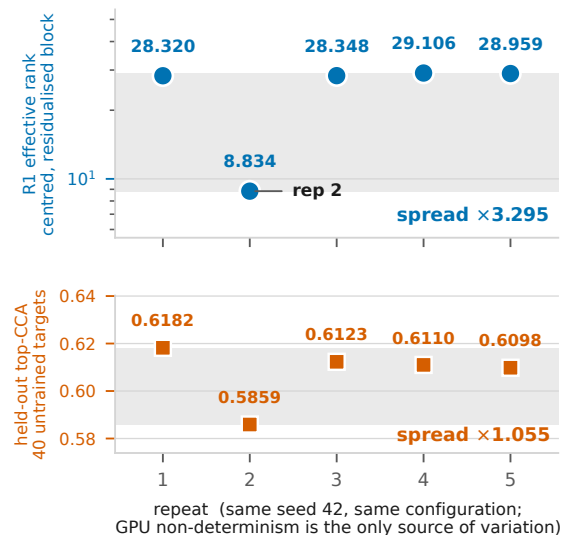
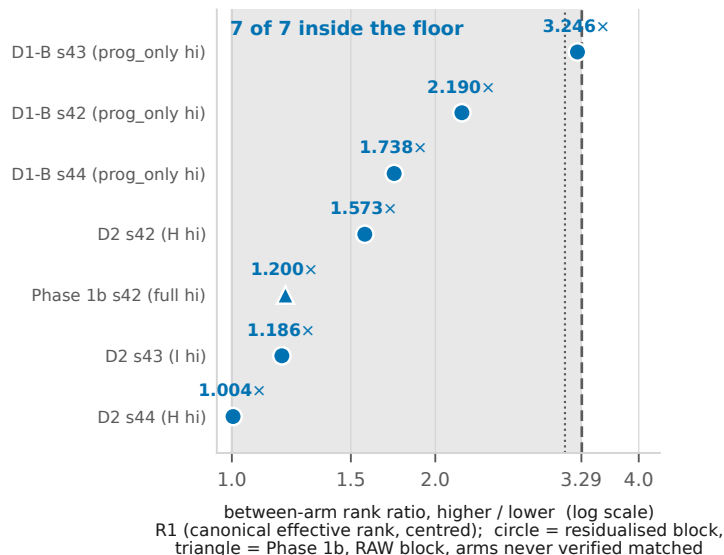


(a) The envelope, measured: 5 identical retrains



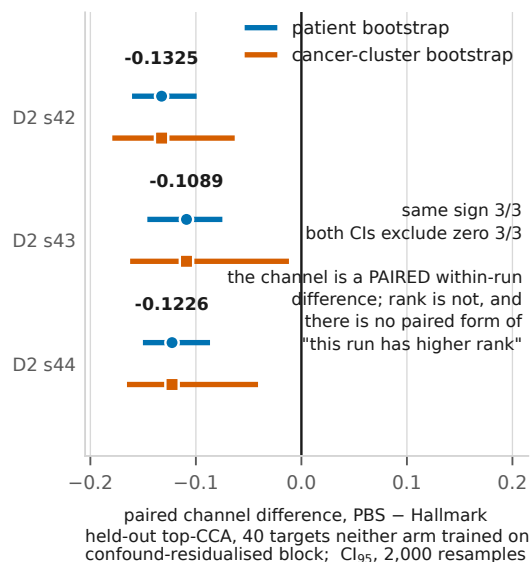
rank $\times 3.295$ against channel $\times 1.055$, on identical inputs.
BIMODAL, not a spread: four repeats agree to 2%, rep 2 lands at a third (rank $\div 3.29$, channel -5%). Reproducible $\sim 80\%$ of the time and catastrophically not $\sim 20\%$ of the time. A FLOOR: programme_only is the stable arm and the seed is held fixed, so both understate the truth.

(b) The seven between-arm differences, against it

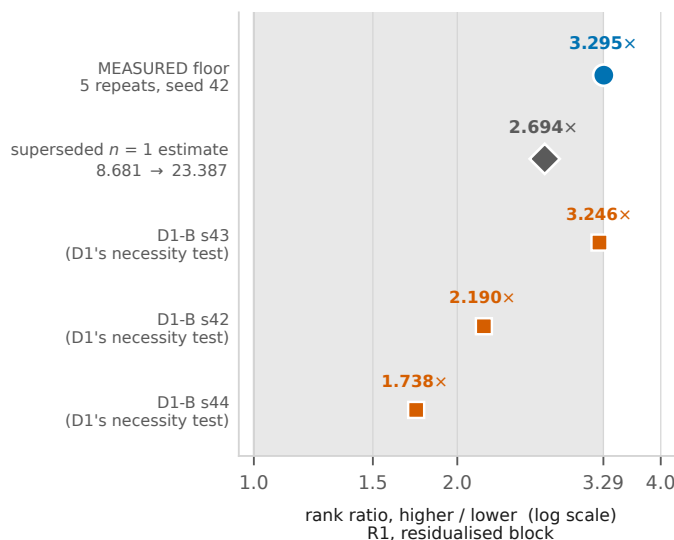


shaded: the measured retraining FLOOR, 1.00-3.295x (residualised).
Dotted line: the RAW-block floor, 3.111x - the Phase 1b point is judged against that one, not against the residualised floor.
Five repeats, one arm, one seed: a floor, not a bound.

(c) The asymmetry, beside it



(d) What the floor replaced, and what it costs us



3 of 3 D1 rank ratios are inside the floor, so D1 is UNINFORMATIVE about rank in either direction. The necessity result is NOT refuted: programme_only still wins the channel 3/3 with patient CIs excluding zero (F6), and it stays at full prominence flagged as unresolvable rather than deleted. The count moved 6/7 to 7/7 IN OUR FAVOUR - and rests on one measurement, five repeats, one arm, one seed. Reported with the scepticism a result against us would get.

F1. The envelope of panel (a) is FIVE identical retrains of ONE configuration at ONE seed on ONE stack (D1 'programme_only', seed 42, GPU non-determinism the only source of variation), read out by v2/research/rebase/d1_envelope_readout.py, which imports every statistic from calibra. It is a FLOOR twice over: programme_only is this project's stable arm, and same-seed repeats exclude seed variation entirely. It cannot separate rank-specific variance from stack non-determinism, architecture or schedule. Statistic R1 (Roy & Vetterli order-1, centred), block confound-residualised for every D2 and D1-B point and RAW for the Phase 1b point, which is judged against the RAW-block floor of 3.111x rather than the residualised 3.295x - and the Phase 1b arms were never verified matched (draft 3.4). Panel (b)'s x-axis is a ratio of two R1 values within one comparison; no two rank statistics share an axis anywhere in this figure. The claim does not rest on this figure alone: F2 reaches the same conclusion from 8 within-arm degrees of freedom without using the floor at all. Panels (b) and (c) sit side by side because the channel is quoted as a paired within-run difference and rank is not - there is no paired form of "this run has higher rank".